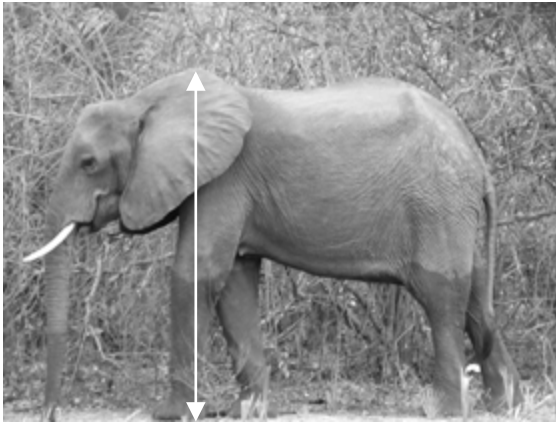


- 8 The photographs show two mammals, an elephant and a mouse.



Magnification $\times 0.02$

- (a) The height of a mouse is 3 cm.

Calculate how many times taller an elephant is than a mouse.

Use the white line drawn on the photograph of the elephant to calculate this value.

(2)

Answer



(b) The respiratory system of an elephant is different from that of other mammals.

The lungs are attached to the chest cavity wall and diaphragm by collagen fibres.

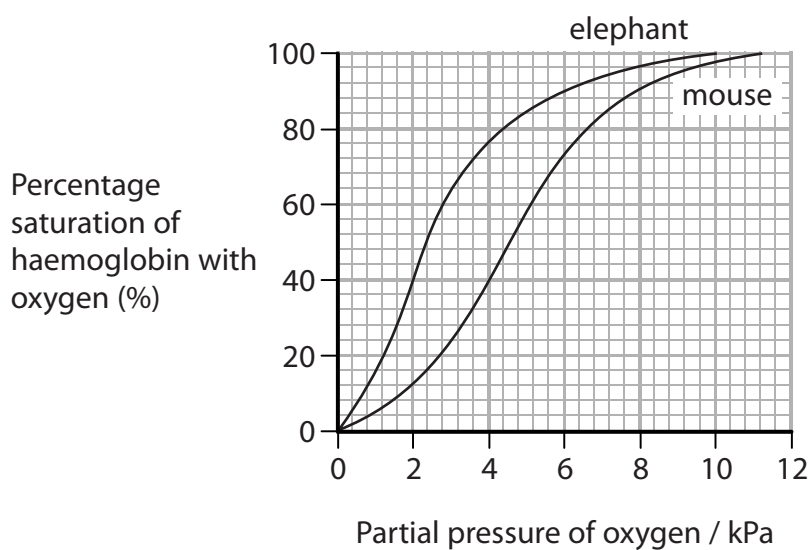
Describe how the lungs of an elephant are adapted for gas exchange.

(3)



- *(c) Graph 1 shows the oxygen dissociation curve of haemoglobin for a mouse and for an elephant.

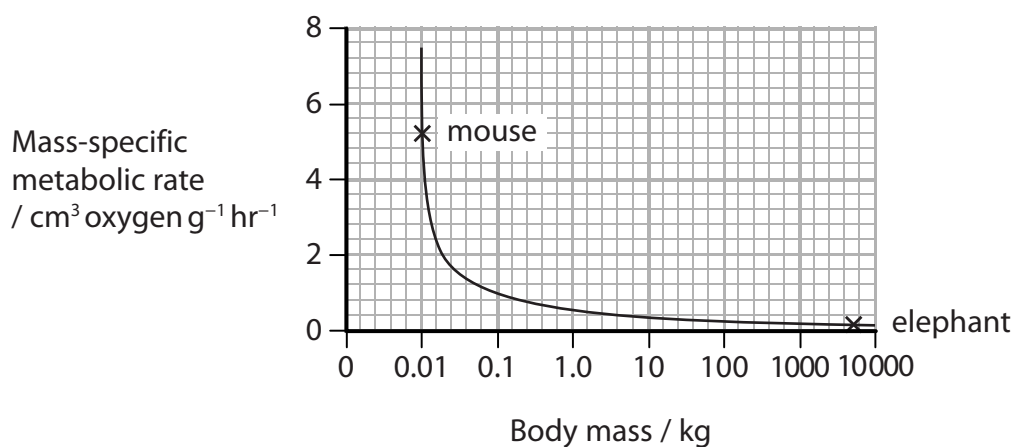
Graph 1



Graph 2 shows the mass-specific metabolic rate for a mouse and for an elephant.

Mass-specific metabolic rate is a measure of how much oxygen is needed for chemical reactions per gram of body tissue.

Graph 2



Explain the difference in the oxygen dissociation curves of haemoglobin for a mouse and for an elephant.

Use the information in both graphs to support your answer.

(6)

(Total for Question 8 = 11 marks)

TOTAL FOR PAPER = 80 MARKS



P 6 0 7 8 8 A 0 2 9 3 2